

JAWAHARLAL NEHRU UNIVERSITY
SCHOOL OF COMPUTER AND SYSTEMS SCIENCES

Mid-Term Exam
M.CA. (Winter Semester-2023)

Course Name: Computer Architecture
Course Code: CS 106

Maximum Marks: 40
Total Time: 2.0 hours

Q1. What is the difference between RISC and CISC type of Computer Architecture? Three enhancements with following speedups are proposed for a new computer:

Speedup₁=25; Speedup₂=15; Speedup₃=10; only one enhancement is useable at time

- (a) If enhancement 1 and 2 are each usable for 22% of the time, what fraction of the time must enhancement 3 be used to achieve overall speed up of 15?
- (b) Assume the enhancement can be used 20%, 35% and 15% of the time for enhancement 1, 2, 3 respectively. For what fraction of the reduced execution time no enhancement is used?

(3+7)

Q2. (a) What are the major performance parameters in designing computer architecture?

b) What are various types of addressing modes in Computer Architecture?

- (c) Suppose a program is running on a machine with the following instruction types, CPI values and the frequencies of occurrence. The CPU designer gives two options i) reduce CPI instruction type A to 1.0 and ii) reduce CPI instruction type B to 1.5. Which one is better?.

Type	CPI	Frequency
A	1.2	50%
B	2.4	30%
C	2.1	20%

(3+2+5
)

Q3. (a) What is the difference between SRAM and DRAM memory cell? Where are they used in the memory hierarchy system and why?

(b) What conclusion can one draw from the following observations in terms of computing performance

Type	CPU A (in Sec.)	CPU B (in Sec.)	CPU C (in Sec.)
Program A	60	40	30
Program B	510	220	90
Total Time	570	260	120

Q4. (a) What is the difference between Access time and Latency in a memory system? What is the difference Associative Mapping N Set Associative Mapping?

(b) Consider a two level memory hierarchy with separate instruction and data cache in level 1 and level 2 respectively. The main memory is at level 3.

Calculate the average access time for the following parameters:

- i) Clock cycle is 1 ns
- ii) Miss cycle is 10 clock cycle
- iii) 2 percent of instruction is not found in Instruction cache
- iv) 9 percent of instruction is not found in Data cache
- v) 25 percent of the total memory access for the data
- vi) Cache Access Time (including hit detection) is 1 clock cycle.

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Maximum Marks: 45

Total Time: 2 hour

Q1. a) Which among Harvard and Von Neumann Architecture is preferred and why?.

Three enhancements with following speedups are proposed for new architecture:

$\text{Speedup}_1=30$; $\text{Speedup}_2=20$; $\text{Speedup}_3=15$; only one enhancement is useable at time

(a) If enhancement 1 and 2 are each usable for 18% of the time, what fraction of the time must enhancement 3 be used to achieve overall speed up of 18?.

(b) Assume the enhancement can be used 30%, 40% and 30% of the time for enhancement 1, 2, 3 respectively. For what fraction of the reduced execution time no enhancement is used?.

(4+6)

Q2. (a) Which performance parameters can improve the execution time of the processor by improving the Computer Architecture?.

b) A weather forecasting computation requires 400 billion floating point operations. The problem is processed in a Super computer that can perform 80 megaflops. How long will it take to perform these calculations?

c) Suppose a program is running on a machine with the following instruction types, CPI values and the frequencies of occurrence. The CPU designer gives two options i) reduce CPI instruction type A to 1.0 and ii) reduce CPI instruction type B to 2.1. Which one is worse?

Type	CPI	Frequency
A	1.5	40%
B	2.5	40%
C	2.0	20%

(3+3+4)

Q3. (a) Why there is a need for Memory Hierarchy in Computer Architecture. Explain with important parameters the Memory Hierarchy in Computer System?. How can one increase the relative size in a Memory system?.

b) What is the difference between SRAM and DRAM cell. Explain with their working.

(5+5)

Q4. (a) What are the total external connections in 256X16 Memory system?. What is the 90/10 rule?. What is the difference between Temporal and Spatial locality in memory system?.

(b) Explain the difference between Direct, Full Associative and Set Associative cache mapping. Which one is preferred is preferred and why ?.

(5+3+2)

Q5. (a) Consider a CPU with an average CPI as 1.8. Assume the instruction mix is ALU-35%, Load-15%, Store-20% Branch-30%. Assume the cache miss rate of 5% and miss penalty of 60 cycles ($=t_{MM}$). Calculate the effective CPI for a unified L1 cache.

(5)

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Maximum Marks: 40

Total Time: 2 hour

Q1. a) Explain the working of SRAM cell

b) Explain RISC is better architecture for Parallel computing or Pipeline computing?

c) Write an assembly language program to show RISC is better than CISC Instruction set.

(4+2+4)

Q2. (a) Explain the additional addressing modes in 8086 microprocessors w.r.t. 8085 microprocessor.

(b) Explain Stack pointer and Program counter. (Use Push and Pop statement wherever necessary). Write an assembly language which results in the sum of first 9 Fibonacci numbers.

c) A scientific simulation program has the following execution profile:

- 35 % of the workload is fully parallelizable.
- 35 % of the workload is partially parallelizable across at most 12 processors.
- The remaining 30% is completely serial.

You upgrade the system to a 16-processor machine. Compute the program's overall speedup on the 16-processor machine. Compute the utilization of the 16 processors (i.e., the fraction of total potential processing power used).

(2+5+3)

Q3. (a) How to improve the latency in a Memory System? Which level cache would be better for Temporal and

Spatial locality in memory system and why?

b) Consider a CPU with an average CPI as 2.3. Assume the instruction mix is ALU-25%, Load-20%, Store-25% Branch-30%. Assume the cache miss rate of 10% and miss penalty of 60 cycles ($=t_{MM}$). Calculate the effective CPI for a combined L1 and L2 cache system.

(4+6)

Q4. (a) What is the difference between non linear dynamic and static pipelining technique?

(b) Which of the cache mapping is the best and worst and why?

© Explain the Arithmetic pipeline for performing subtraction. Deduce the reservation Table and determine its MAL?

(2+3+5)

Key

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Q1. a) What is the difference between Computer Organization and Computer Architecture? Which parameters are directly related to processing power of the computer?

b) Amdahl's law is applicable to parallel processing or pipeline processing?

c) Write an assembly language program to show the difference between RISC and CISC Instruction set.
(4+2+4)

Q2. a) What are different types of addressing modes in 8086 microprocessor

b) A weather forecasting computation requires 40 Trillion floating point operations. The problem is processed in a Super computer that can perform 80 megaflops. How long will it take to perform these calculations?

c) A scientific simulation program has the following execution profile:

- 50% of the workload is fully parallelizable.
- 30% of the workload is partially parallelizable across at most 8 processors.
- The remaining 20% is completely serial.

You upgrade the system to a 32-processor machine. Compute the program's overall speedup on the 32-processor machine. Compute the utilization of the 32 processors (i.e., the fraction of total potential processing power used).
(3+3+4)

Q3. (a) Explain the important parameters that defines the performance of a Memory System? How can one increase the relative size in a Memory system? What is the 90/10 rule? What is the difference between Temporal and Spatial locality in memory system?

b) Consider a CPU with an average CPI as 2.0. Assume the instruction mix is ALU-35%, Load-15%, Store-20% Branch-30%. Assume the cache miss rate of 5% and miss penalty of 60 cycles ($=t_{\text{miss}}$). Calculate the effective CPI for a unified L1 cache.

(5+5)

Q4 (a) What are the total external connections in 256×32 Memory system?

(b) Explain the difference between Direct, Full Associative and Set Associative cache mapping. Which one is preferred and why?

(c) Explain the Arithmetic pipeline for performing multiplication. Deduce the reservation Table and determine its MAL?

(2+3+5)

25×100